Models and AI Methods

Task	Technology	Role
Ingestion	PyMuPDF	Extract text, blocks, pages
Cleaning / chunking	Python	Normalize PDF text and section chunks
Embedding	all-MiniLM-L6-v2 via Sentence Transformers	Pretrained text-to-vector inference
Retrieval	ChromaDB	Similarity-ranked evidence search
Answer generation	Google Gemini	Grounded natural-language output
Test provider	MockLLMProvider	Repeatable offline responses, no API costs
Citation verification	Python CitationVerifier	Validate retrieved citation IDs
Entity extraction	Regex/table rules, optional LLM fallback	SWC/Port/Interface records

No new transformer was trained or fine-tuned. No PCA/t-SNE/UMAP dimensionality reduction is implemented. A sentence embedding is a learned vector representation, not PCA.

Model ID discrepancy: config/settings.py (../config/settings.py) defaults to gemini-3-flash-preview, whereas root .env.example (../.env.example) specifies gemini-2.5-flash. The real effective model depends on the private .env used for the specific run; do not report a model that was not verified.

Responsible interpretation: a valid citation identifier confirms source provenance, not that each claim is factually supported. A human reviewer checks the quoted text.